

Object-Oriented Programming (CS112)

Lab Task-9

[CLO-2, Taxonomy Level-C3, GA3]

Course: BSSE-II

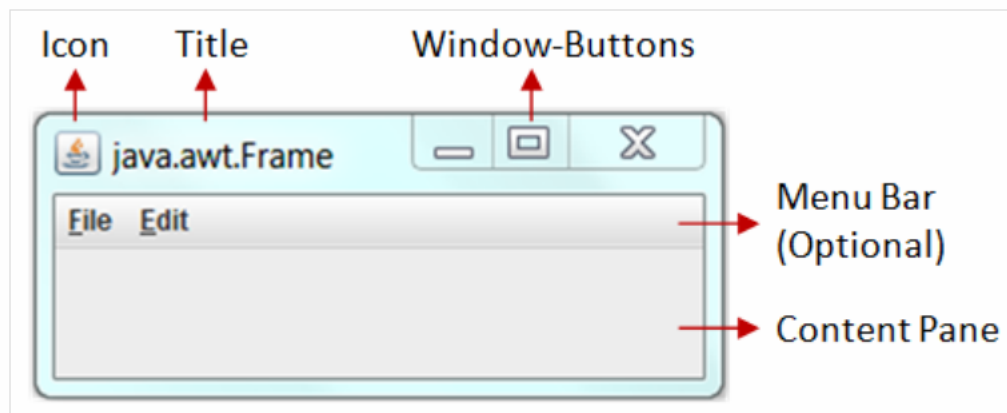
Semester:2nd (Spring 2025)

Instructions

Plagiarism, copy & paste material will lead to the cancellation of your assignment. The input values should be different, don't copy the solution from external source as the plagiarism will be tested through Turnitin and the plagiarized assignments will lead to marks' deduction.

Objectives:

- Java frames
- Buttons
- Layouts
- Dialogboxes



Activity

Showing a dialog box

```
JOptionPane.showMessageDialog(null, "This is an information message.", "Info",  
JOptionPane.INFORMATION_MESSAGE);
```

1. **Showing prompt to enter user name in a dialog box, when user entered: Message on console should be displayed.**

```
String name = JOptionPane.showInputDialog("Enter your name:");  
System.out.println("User entered: " + name);
```

2. **Activity (Grid Layout)**

GridLayout in Java Swing is a layout manager that arranges components in a grid of rows and columns — like a table.

All cells in the grid are equal size.

Components are added left to right, top to bottom.

You define the number of rows and columns when you create the layout.

Syntax:

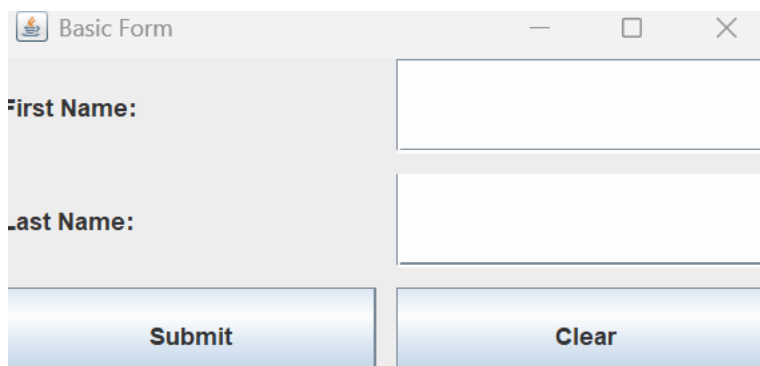
```
frame.setLayout(new GridLayout(3, 2, 10, 10)); // 3 rows, 2 columns
```

```
new GridLayout(rows, columns, hgap, vgap)
```

or

```
new GridLayout(rows, columns)
```

Create the following GUI with the Grid layout



Solution:

```
import javax.swing.*;
import java.awt.*;

public class Hello {

    public static void main(String[] args) {

        // Create a frame

        JFrame frame = new JFrame("Basic Form");

        frame.setSize(400, 200);

        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        frame.setLayout(new GridLayout(3, 2, 10, 10)); // 3 rows, 2 columns
```

```

JLabel label1 = new JLabel("First Name:");
JTextField textField1 = new JTextField();

JLabel label2 = new JLabel("Last Name:");
JTextField textField2 = new JTextField();

JButton button1 = new JButton("Submit");
JButton button2 = new JButton("Clear");

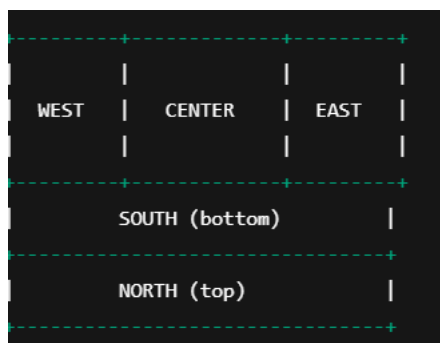
frame.add(label1);
frame.add(textField1);
frame.add(label2);
frame.add(textField2);
frame.add(button1);
frame.add(button2);

frame.setVisible(true);
}
}

```

3. Border Layout

BorderLayout is another layout manager in Java Swing that arranges components in five regions:



Syntax: `frame.setLayout(new BorderLayout());`

Adding components: `frame.add(component, BorderLayout.NORTH);`

`frame.add(component, BorderLayout.SOUTH);`

```
frame.add(component, BorderLayout.EAST);  
frame.add(component, BorderLayout.WEST);  
frame.add(component, BorderLayout.CENTER);
```

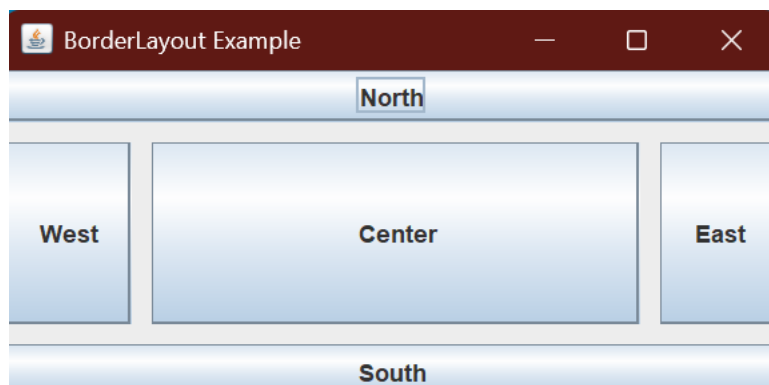
Example:

```
JButton b1 = new JButton("North");  
JButton b2 = new JButton("South");  
JButton b3 = new JButton("East");  
JButton b4 = new JButton("West");  
JButton b5 = new JButton("Center");
```

```
frame.setLayout(new BorderLayout());
```

```
frame.add(b1, BorderLayout.NORTH);  
frame.add(b2, BorderLayout.SOUTH);  
frame.add(b3, BorderLayout.EAST);  
frame.add(b4, BorderLayout.WEST);  
frame.add(b5, BorderLayout.CENTER);
```

Create the following GUI with Border layout



Solution:

```
import javax.swing.*.*;  
import java.awt.*.*;  
public class Hello {
```

```

public static void main(String[] args) {
    JFrame frame = new JFrame("BorderLayout Example");
    frame.setSize(400, 200);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.setLayout(new BorderLayout(10, 10)); // Using BorderLayout

    // Create buttons
    JButton buttonNorth = new JButton("North");
    JButton buttonSouth = new JButton("South");
    JButton buttonEast = new JButton("East");
    JButton buttonWest = new JButton("West");
    JButton buttonCenter = new JButton("Center");

    // Add buttons to the frame in different regions
    frame.add(buttonNorth, BorderLayout.NORTH); // Top region
    frame.add(buttonSouth, BorderLayout.SOUTH); // Bottom region
    frame.add(buttonEast, BorderLayout.EAST); // Right region
    frame.add(buttonWest, BorderLayout.WEST); // Left region
    frame.add(buttonCenter, BorderLayout.CENTER); // Center region

    // Make the frame visible
    frame.setVisible(true);
}
}

```

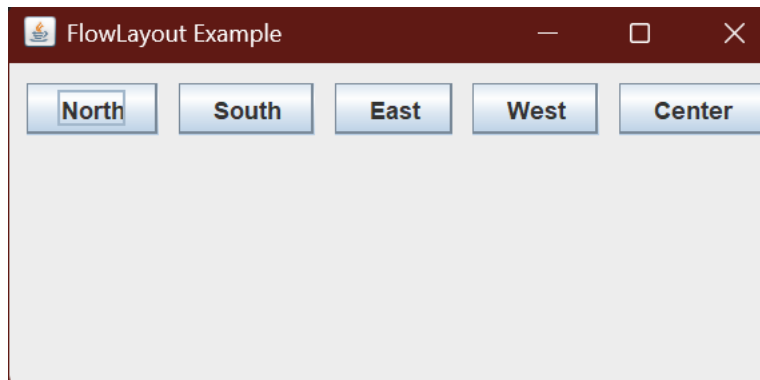
4. Flow Layout

FlowLayout

- **Description:** Arranges components in a **left-to-right flow** (like text in a paragraph), and if the space is full, it moves components to the next line.
- **Use Case:** Great for small apps where components don't need strict alignment.

Syntax:

```
frame.setLayout(new FlowLayout());
```



Solution:

```
import javax.swing.*;
import java.awt.*;

public class Hello {
    public static void main(String[] args) {

        JFrame frame = new JFrame("FlowLayout Example");
        frame.setSize(400, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new FlowLayout(FlowLayout.CENTER, 10, 10)); // Using FlowLayout
        with centered alignment and spacing

        // Create buttons
        JButton buttonNorth = new JButton("North");
        JButton buttonSouth = new JButton("South");
        JButton buttonEast = new JButton("East");
        JButton buttonWest = new JButton("West");
        JButton buttonCenter = new JButton("Center");

        // Add buttons to the frame
        frame.add(buttonNorth);
        frame.add(buttonSouth);
        frame.add(buttonEast);
        frame.add(buttonWest);
        frame.add(buttonCenter);
```

```

        // Make the frame visible
        frame.setVisible(true);
    }
}

```

LAB TASK-9

1. Create the following GUI with title “MyFrame” having the following components.

The GUI for "MyFrame" consists of a light gray background. It features two input fields: the top one is labeled "Enter First Number" and the bottom one is labeled "Enter Second Number". Below these fields, there are four buttons arranged horizontally, each with a mathematical operator: "+", "-", "*", and "/".

2. Create the following GUI

The GUI has a light gray background. It contains two input fields: "First Name:" and "Last Name:". Below these fields, there are three buttons in a row: "Clear", "Help", and "Submit". At the bottom of the window, there is a single button labeled "Exit".

3. Create the following GUI

The GUI has a light gray background. It features two input fields: "First Name:" and "Last Name:". Below the "First Name:" field, there is an empty rectangular box. To the right of this box, there are two buttons: "Submit" and "Clear". Below these, there are three buttons in a row: "Reset", "Exit", and "Help".